# Radio Wave Propagation

112-BOLCEW07

# Concrete Experience



**Please Answer the Following Questions**

-Do you Have WIFI in your House?

-Dose your WIFI have 2 different signals to connect to?

-What are they and what do the numbers stand for?

-Which signal get's you faster download speeds?

-Which signal reaches further?

-Do you loose signal anywhere in your house and if so, where?

# Learning Step Activities



- Define the basics of radio wave propagation
- Define the types of Radio Frequency (RF) propagation paths
- Define Earth's Atmosphere and effects on propagation
- Define types of wave degradation
- Define the maximum, optimum, and lowest usable frequencies of radio waves
- Define weather considerations with radio wave propagation
- Identify electromagnetic interference, man-made/natural interference, and ionospheric disturbances affect radio wave propagation

# Wave propagation on Earth



**Free Space**

-Not many obstacles

-Little to no weather affects

-EM Waves will continue moving infinitely until it hits something.

-EM waves from stars can easily move trillions of miles before absorption.

-Typically, EM Waves will spread out radially in all directions from a source.

*VS*

**On Earth**

-Lots of obstacles (tress, hills, cities, etc…)

-Lots of weather effects

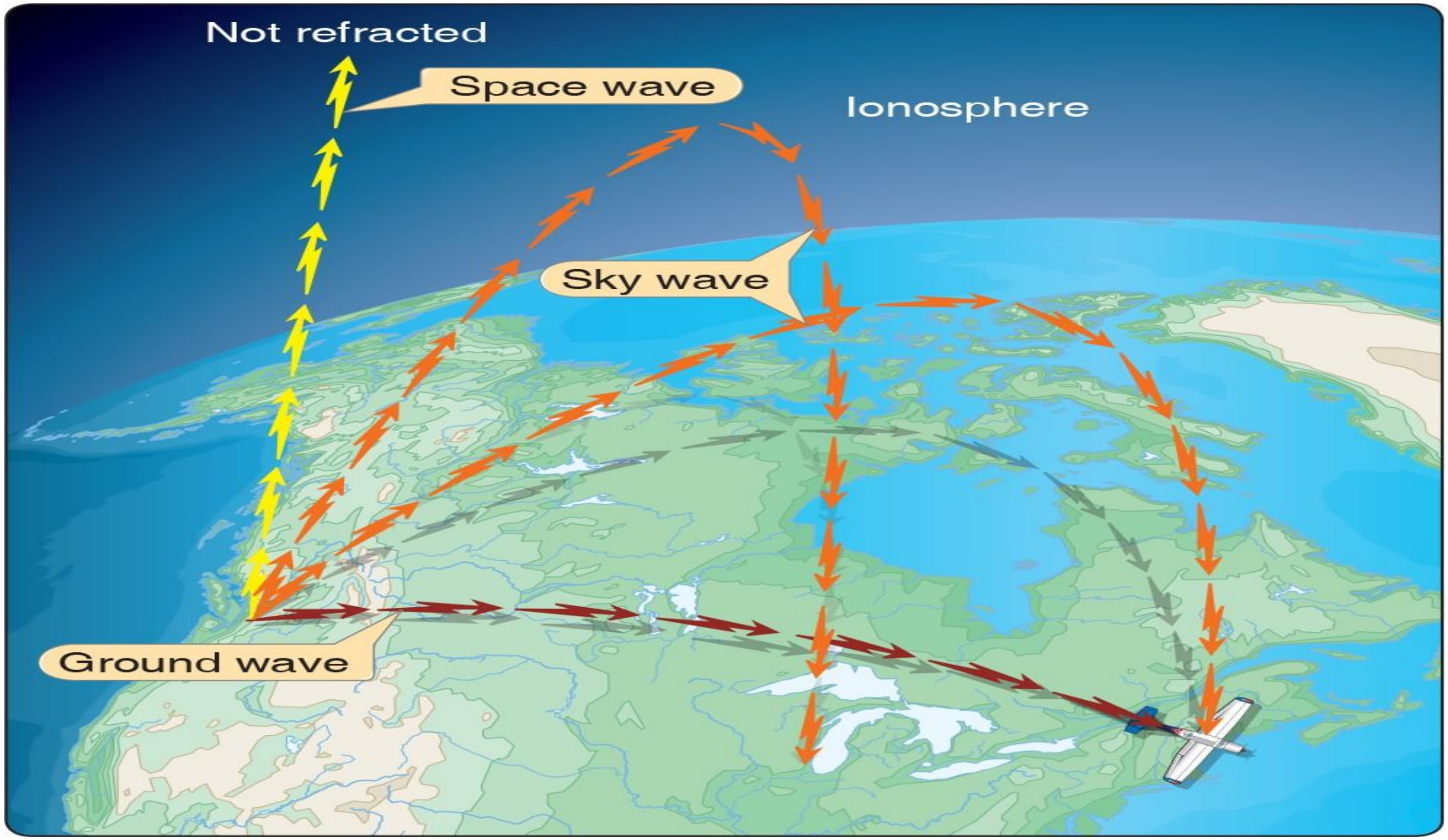
- Layers of the atmosphere can become charged and interfere with EM waves

-Curvature of the earth significantly reduces LOS for waves (about 6 miles near the earth's surface)

-Man made and natural occurring sources of interference (power lines and lightning)

# Types of Radio Frequency (RF) Propagation Paths





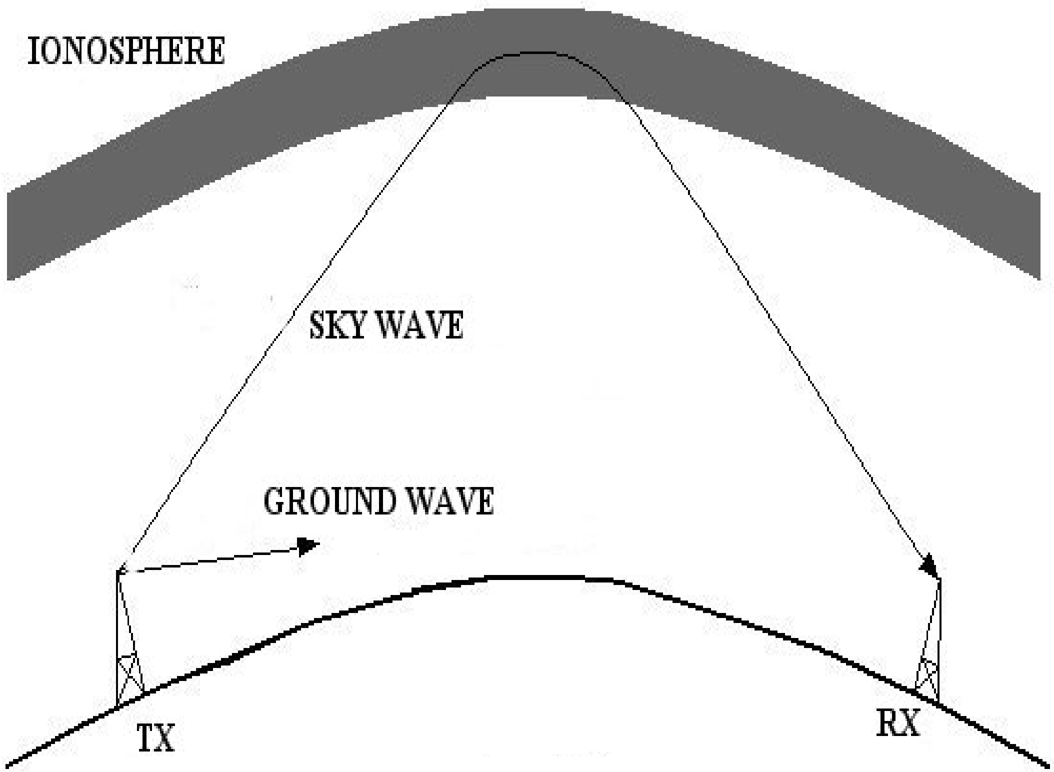
# Radio Wave Transmission Paths or Methods



**Radio Waves on Earth have two Transmission Paths:**

- Ground waves
- Sky waves

- Radio waves that leave Earth
- Space Wave



# Ground Wave

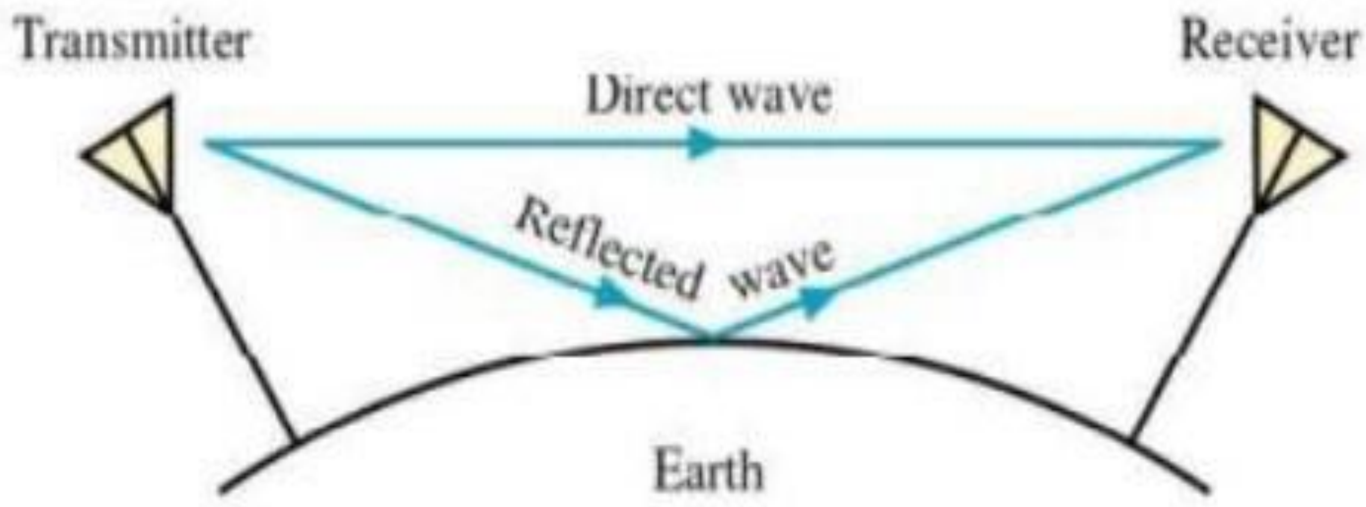


Ground waves:

- These are known as Surface, Direct; and Ground Reflected Waves

These are affected by:

- Terrain
- Earth's electrical characteristics
- Diffraction (bending) along the Earth's curvature

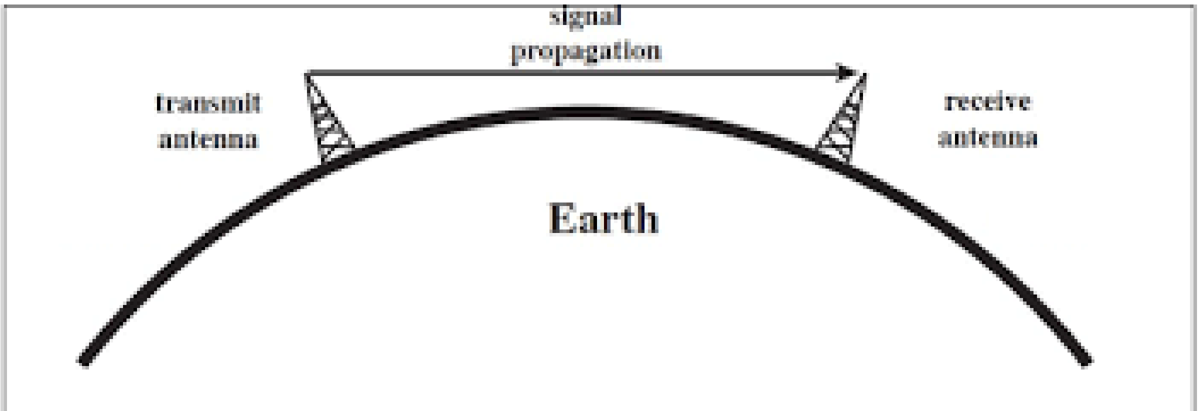


# Direct Ground Waves



Have a **LOS Propagation** path that:

- travels in a direct path from the source to the receiver
- **Extremely affected by earth curvature, terrain and large natural obstacles ie (mountains)**
- Can travel through many non-metallic objects and go through walls (depending on the frequency)




Line-of-sight (LOS) propagation (above 30 MHz)

# Surface Wave (1 of 2)



Surface Wave travel along the surface of the Earth

- These are further attenuated by the Earth’s magnetism
  - Voltage or Earth’s shape and conductivity along the transmission path
- Can follow because of the process of **diffraction**
- Slightly longer line of sight than direct waves.

Attenuation that a surface wave undergoes because of induced voltage also depends on the electrical properties of the terrain over which the wave travels

# Surface Wave (2 of 2)



Surface Wave (cont.):

- Surface Conductivity

| *SURFACE* | *RELATIVE CONDUCTIVITY* |
|---|---|
| Sea water | Good |
| Flat, loamy soil | Fair |
| Large bodies of fresh water | Fair |
| Rocky terrain | Poor |
| Desert | Poor |
| Jungle | Unusable |

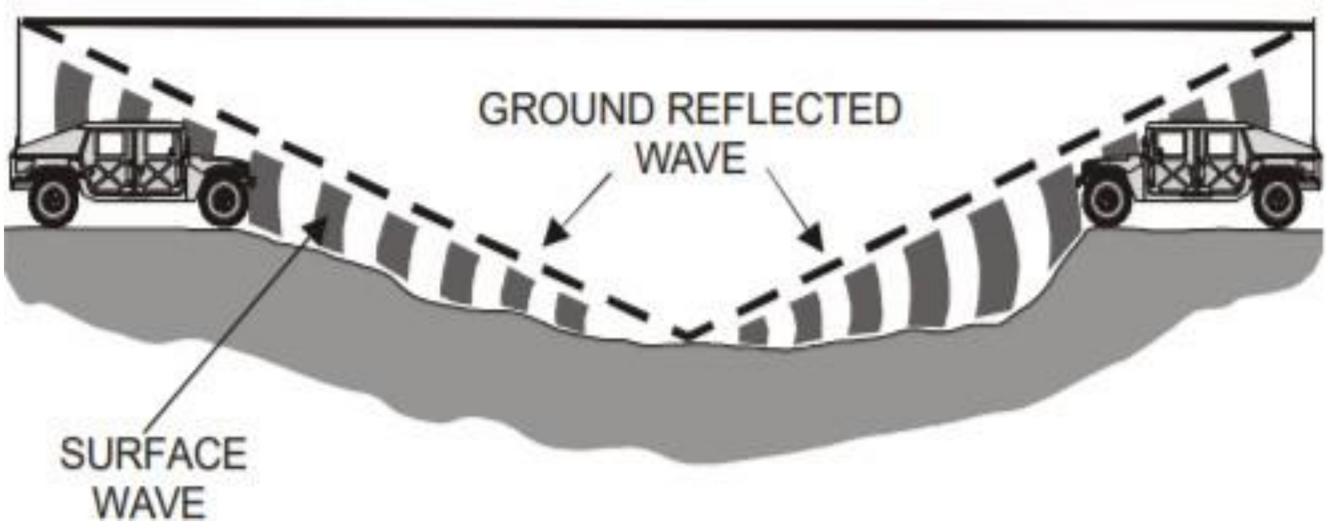
# Ground Reflected Wave

Ground wave that is **reflected** off the earth:

-Can help Ground waves “move over” a tall obstacle.

-Generally, has a shorter coverage are for receivers on the ground.

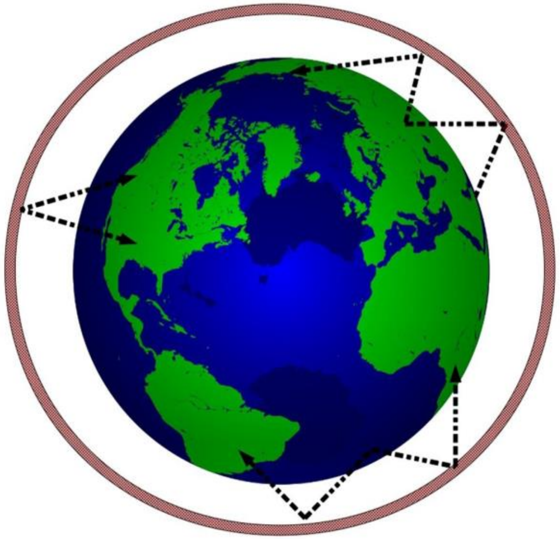
-May lead to unwanted reception



# Sky Wave

Sky Wave:

- Often called the Ionospheric Wave
- Radiated in an upward direction and returned to Earth
- Distant location because of refraction from the Ionosphere

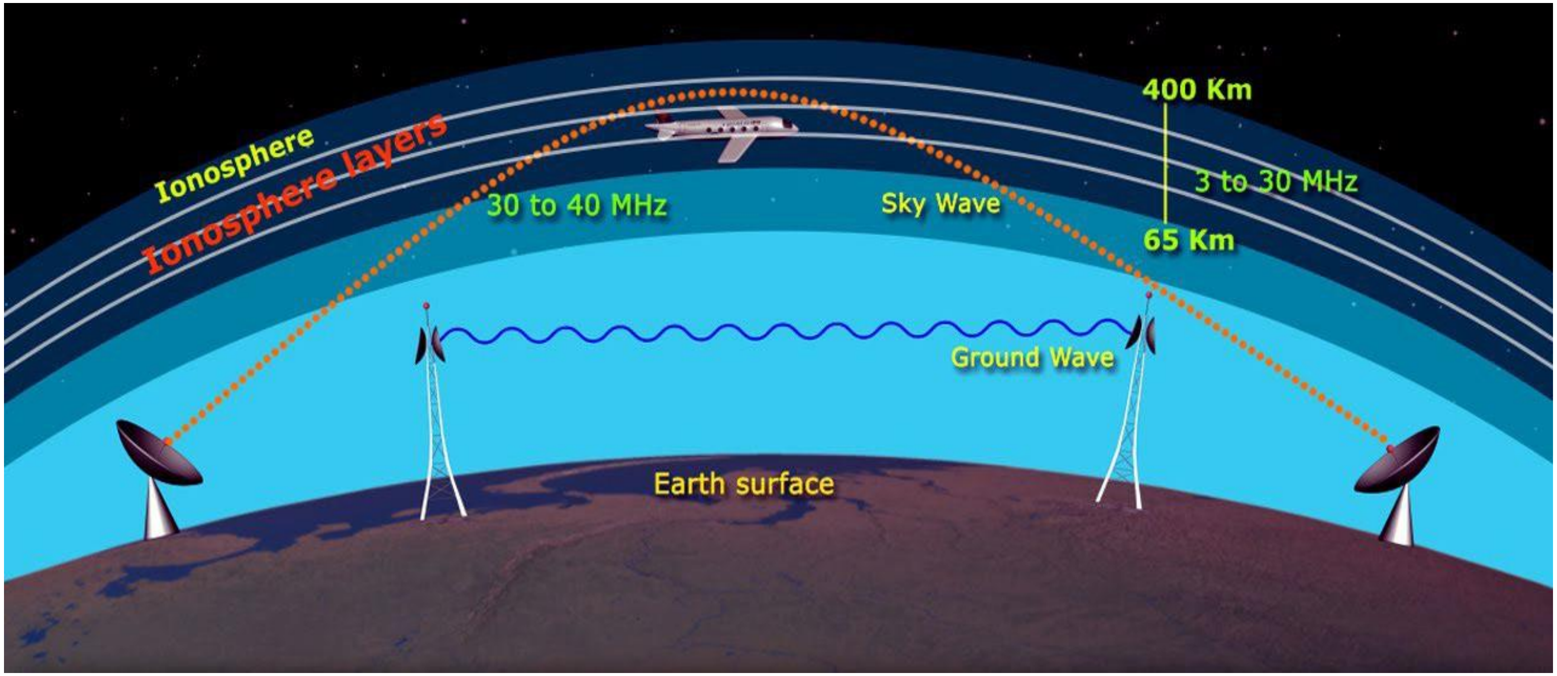


- This form of propagation can travel great distances
- Not effected by the Earth's surface (Ground waves)

# Earth’s Atmosphere and Effects on Propagation



- How is Earths Atmosphere divided?
- How does the Atmosphere affect propagation?

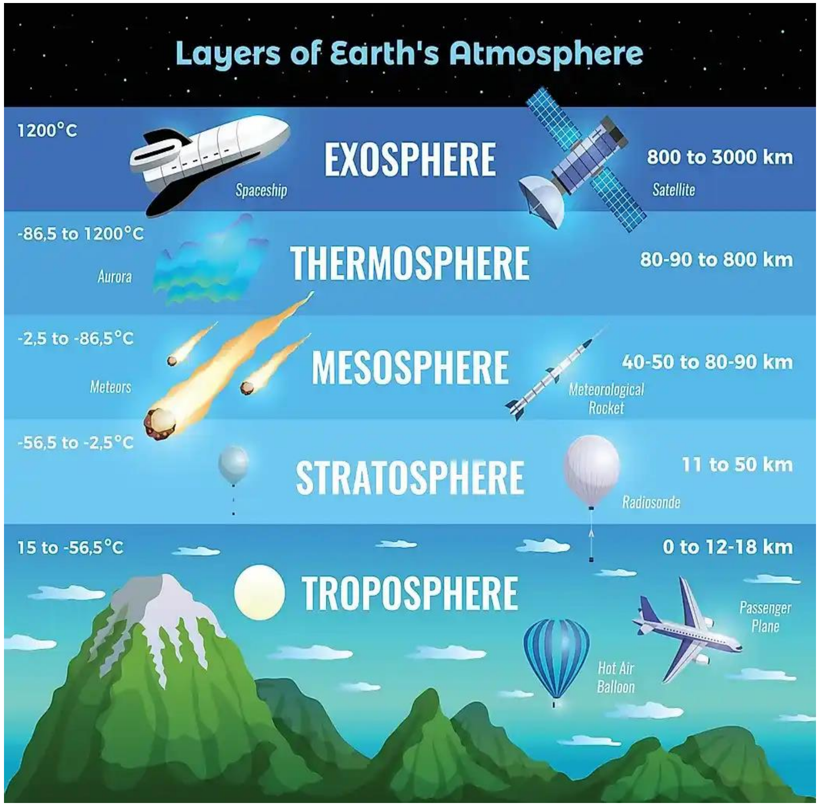


# Earth's Atmosphere



The atmosphere surrounding the Earth is divided into several layers, each possessing unique characteristics.  The most relevant layers:

- Troposphere
- Stratosphere
- Ionosphere (in the Meso and Thermosphere)



# Troposphere

Troposphere is:

- Where weather variations occur
- Big effect on Radio Wave propagation

# Stratosphere



Stratosphere is:

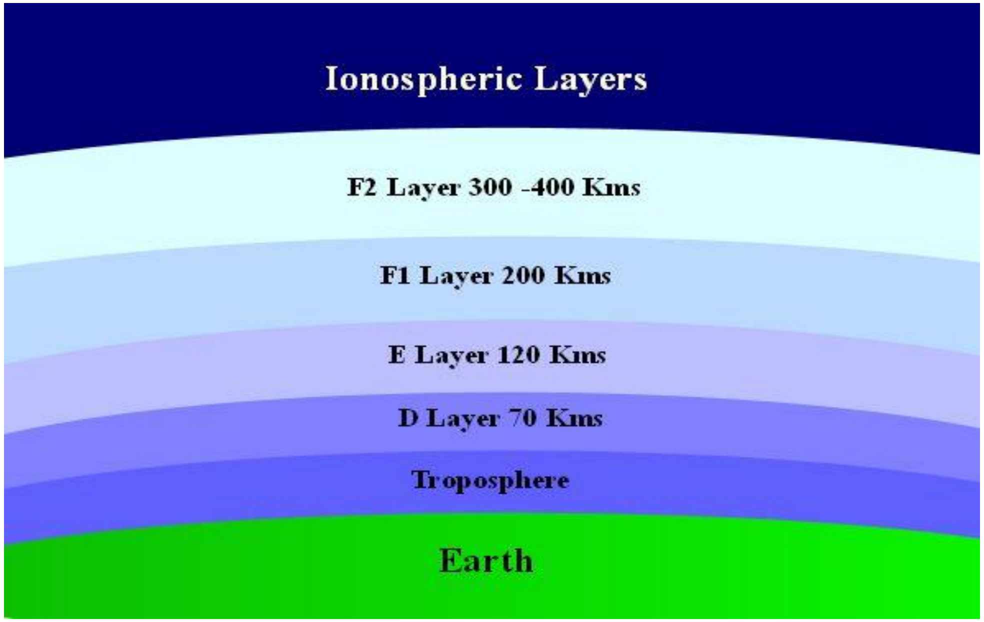
- Located between the Troposphere and Ionosphere
- Temperature is constant
- Has little to no effect on Radio Waves
- An extremely calm region

# Ionosphere



Ionosphere is:

- Part of the Meso and Thermosphere that is ionized by solar radiation
- Changes size and properties during different times and conditions
- Contains four layers of electrically charged ions during the day and two during at night
- Refracts radio waves back to Earth



# Types of Wave Degradation



# Wave Degradation

Wave degradation is anything that causes a signal loss. Several types of degradation are:

- Weather
- Transmission Loss
- Electromagnetic Interference (EMI)

# Transmission Losses



All radio waves lose energy before arriving at the receiver.

Two types of losses:

- Ground Reflection
- Free Space

The combined effects of:

- Absorption
- Ground Reflection Loss
- Free Space Loss

Accounts for most of the energy losses of radio transmissions propagated by the ionosphere

# Absorption



When a radio wave passes through an object or medium some of its energy is absorbed

Absorption results in the loss of energy, of a radio wave, and has a pronounced effect

- Both the strength of received signals and the ability to communicate long distances
- Greatest loss of radio wave energy is due to absorption
- Ground waves lose energy because of ground induced voltage
- Sky wave absorption occurs in high density layers of the ionosphere
- Absorption is more significant at higher frequencies

# Ground Reflection Loss



Amount of energy lost depends on:

- Frequency
- Angle of  incidence
- Ground irregularities
- Electrical conductivity

RF energy is lost each time the radio wave is reflected from the earth's surface.

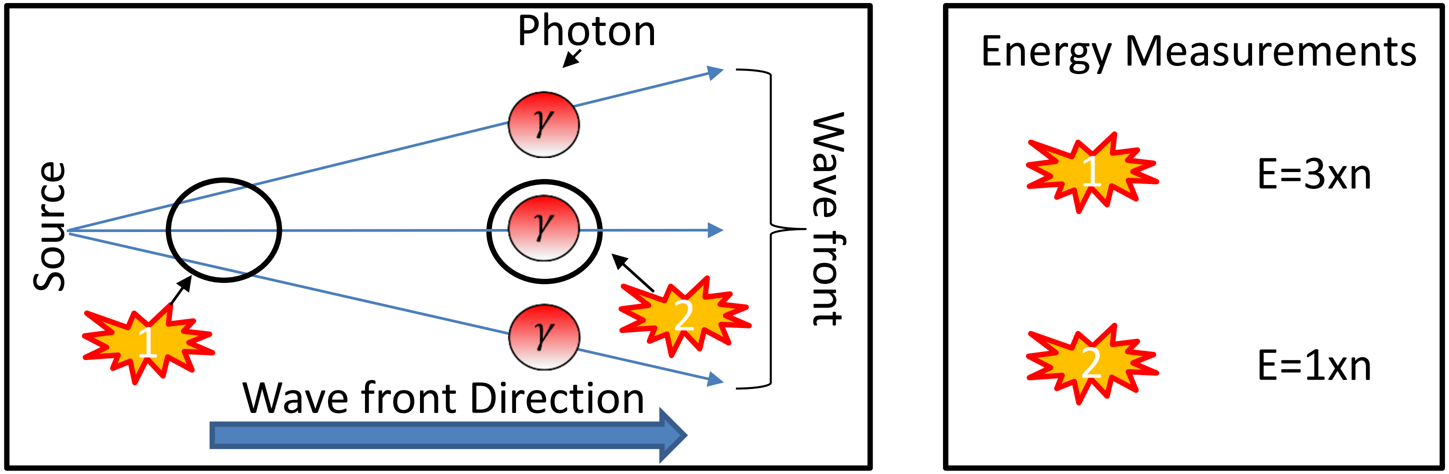
# Free Space Loss

Free Space Loss is the loss in signal strength that would result from a line-of-sight path through free space (usually air)

Major free space loss is due to the spreading out of the wave front (like a flashlight):

- Energy decreases as distance increases

## Wave Front Spreading



# Weather Considerations with Radio Wave Propagation



# Weather



Weather plays a factor in Radio Wave Propagation

- Wind
- Air Temperature
  - Can cause ducting
- Precipitation (rain/fog/snow)

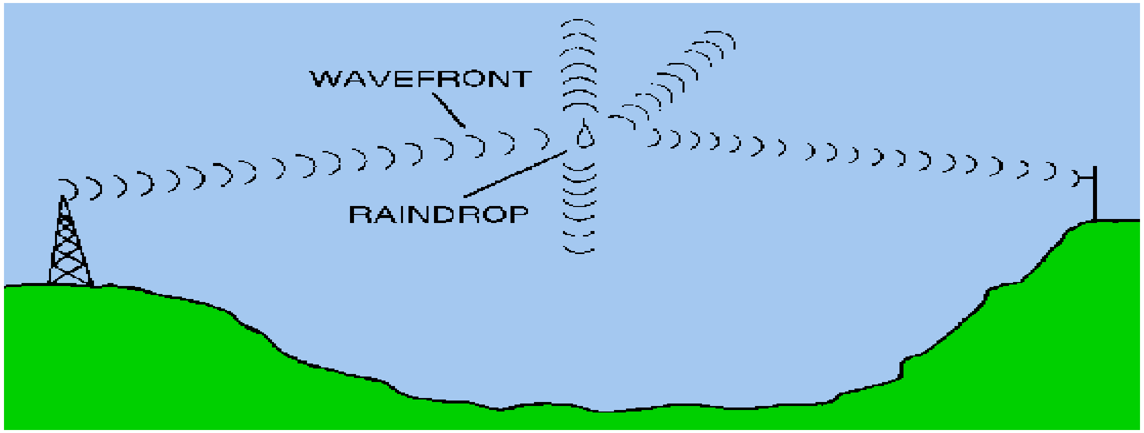
All of these can “COMBINE” in many ways to impact radio waves

- Example: Extreme case it will go beyond the intended destination by hundreds of miles
- Opposite Combination: The radio wave could not be heard on a satisfactory path or trajectory

# Rain

Rain

- Attenuates waves by scattering
  - The effect of scattering increase significantly has the frequency increases
- Absorbs energy



# Fog

Fog

- May be considered as a  form of rain
- Attenuation is determined by quantity of water suspended in air
- Can cause serious  absorption at frequencies greater than 2 GHz

# Snow



Snow effects are difficult to compute

- Various sizes and shapes
- Rain is 8 times more dense than snow
- Attenuation from snow is less than rain moving at the same rate

Hail

- Attenuation determined by stone size
- Chance of scattering, absorption, or attenuation is considerably less than rain or fog

# Other Phenomenon

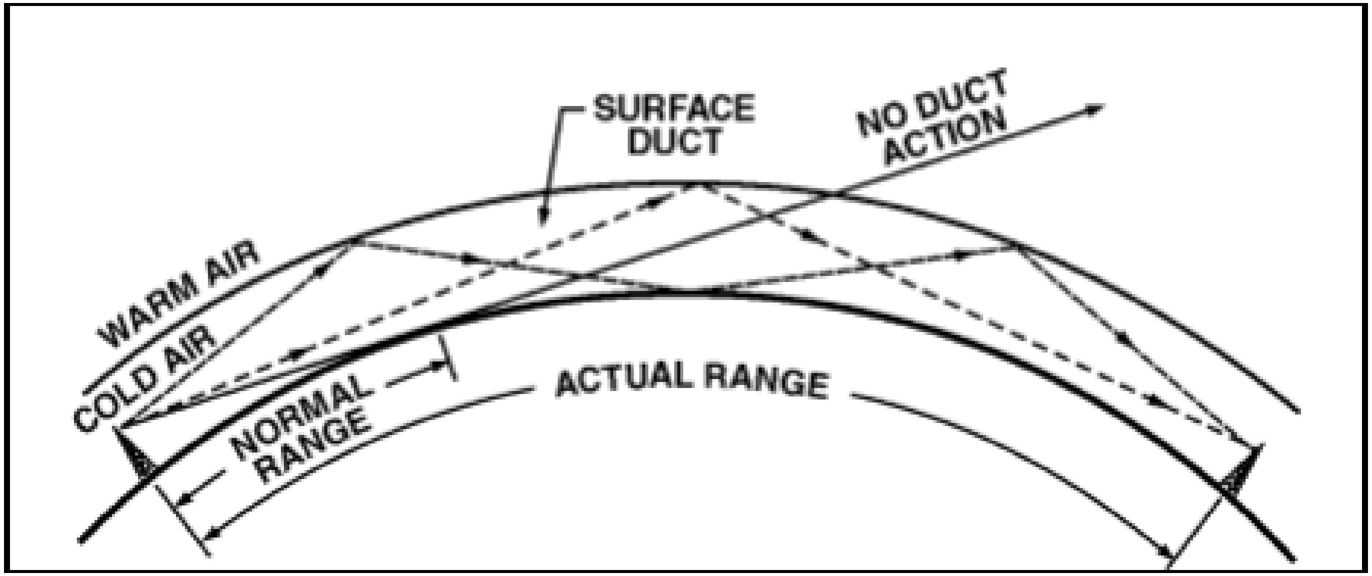


**Temperature Inversion (Ducting)**

Commonly referred to as “Ducting”

Radio waves get “trapped” between differing air temperatures

Can achieve great distances; Over-the-Horizon (OTH)



# Other Phenomenon



**Tropospheric Scattering:** This scattering mode of propagation enables VHF and UHF signals:

- Transmitted far beyond the normal line-of-sight
- Used for Over The Horizon (OTH) reception
- May reach up to 500 miles

# Questions?

# EMI, Man-made/Natural Interference, and Ionospheric Disturbances that Affect Radio Wave Propagation



# Electromagnetic Interference (EMI)



EMI is:

Interference that results in annoying or impossible operating conditions

Sources of EMI are both man made and natural

# Electromagnetic Interference (EMI) Man-Made



Man-Made Interference comes from:

- Communication transmitters
- Electrical Devices
- Ignition systems
- Motors
- Switches

The intensity of man-made vary throughout the day

- Lowers in night

Higher interference noted near industrial areas

# Electromagnetic Interference (EMI) – Natural



Natural Interference is defined by:

- Generated by natural phenomena (occurrence)
- Energy released is transmitted to receiving site in the same manner as radio waves

When the Ionosphere conditions are favorable:

- Long distance propagation can be affected by natural Interference

# Controlling EMI

Electromagnetic interference can be reduced or eliminated by:

- Cutting transmitting antennas to the correct frequency
- Limiting bandwidth
- Using electronic filtering networks and shielding

Radiated EMI during transmission can be controlled by:

- Physical separation of the transmitting and receiving antennas
- Use of directional antennas
- Limiting antenna bandwidth

# Summary

In summary, you learned to:

- Define Electromagnetic Waves
- Compare Electromagnetic Waves and normal Waves
- Recognize the key characteristics of an Electromagnetic wave
- Calculate Electromagnetic wave measurements
- Define magnetic, electric, electromagnetic fields
- Compare Electromagnetic Waves propagation to Normal waves
- Define the basics of Electromagnetic wave propagation
- Define how waves are polarized
- Define types of wave Interactions
- Understand the doppler effect for Electromagnetic waves

# References



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